YOUNGJUN LEE, PH.D.

CONTACT INFORMATION

Department of Pediatrics, Division of Gastroenterology, Hepatology and Nutrition
Children's Hospital of Philadelphia
Philadelphia, PA, United States
E-mail: leey14@chop.edu
Phone: +1 (517) 220-8178
ORCID: 0009-0001-3915-1389
Google Scholar

RESEARCH INTERESTS

(1) Population and Wearable-Based Behavioral Epidemiology of Physical Activity, Sleep, and Mental Health in Children and Youth, with a Focus on Autism and Developmental Disabilities; (2) Digital Behavioral Health Interventions, Including Immersive Virtual Reality Exercise with Virtual Coaches and Wearable-Informed Approaches

EDUCATION

[2025] **Michigan State University (Ph.D.)** Michigan, United States
Department of Kinesiology, College of Education; Physical Activity in Youth Disabilities Lab
Dissertation: *"Immersive Virtual Reality Exercise to Improve Depressive Symptoms in Young Adults with Autism Spectrum Disorder"*
Advisor: Dr. Janet Hauck

[2022] **Korea University (M.P.E.)** Seoul, Korea
Department of Physical Education, Exercise Nutrition Biochemistry Lab
Dissertation: *"Effect of Physical Activity and Nutritional Behavioral Change on Metabolic Syndrome in Overweight Young Women of Single-person Households"*
Advisor: Dr. Jonghoon Park

[2020] **Korea University (B.P.E.)** Seoul, Korea
Department of Physical Education, College of Education; graduated with Great Honor

PROFESSIONAL EXPERIENCE

[2026 – Present] **Postdoctoral Research Fellow**, Children's Hospital of Philadelphia, Department of Pediatrics, Division of Gastroenterology, Hepatology and Nutrition
Wearable- and health record-based research on physical activity, sleep, and depression in autistic and non-autistic populations (Mentor: Dr. Jonathan A. Mitchell)

[2026 – Present] **Affiliated Postdoctoral Researcher**, University of Pennsylvania, Annenberg School for Communication, Annenberg Extended Reality Lab
Research on wearable sensors, immersive technologies, and social presence in extended reality

REFEREED JOURNAL PUBLICATIONS

Published

[P15] **Lee Y.** Socioeconomic Hardship and Cumulative Adverse Experiences in Relation to Anxiety and Depression in Autistic Children: Exploring the Role of Physical Activity. *Journal of Autism and Developmental Disorders.* 2026. *DOI:* https://doi.org/10.1007/s10803-026-07607-7

[P14] **Lee Y**. Resting Heart Rate Variability and the Persistence of Simulator Sickness During Virtual Reality Exercise in Autistic Young Adults: A Proof-of-Principle Study. *Research in Autism.* 2026;139:203046. *DOI:* https://doi.org/10.1016/j.reia.2026.203046

[P13] **Lee Y**, Ciaccioni S, Stodden D, Hauck J, Lima RA, Pesce C. Physical Activity and Adolescent Mental Health Across 45 Countries: Exploring the Role of Country-Level Perceived Health. *Mental Health and Physical Activity*. 2026;31:100826. *DOI:* https://doi.org/10.1016/j.mhpa.2026.100826

[P12] **Lee Y.** Differential Moderation by Aerobic and Muscle-Strengthening Physical Activity of the Violence Victimization-Suicidal Ideation Association in Korean Adolescents. *Suicide and Life-Threatening Behavior.* 2026; 56:e70134. *DOI:* https://doi.org/10.1111/sltb.70134

[P11] Ciaccioni S, **Lee Y**, Capranica L, Urban A, May R, Massini S, Guidotti F. Building resilience and equity: An umbrella review of evidence for crisis management in grassroots sport. *Societies*. 2026;16(4):118. DOI: https://doi.org/10.3390/soc16040118

[P10] **Lee Y**, Guidotti F, Capranica L, Pesce C, Benzing V, Hauck J, Ciaccioni S. Olympic combat sports and mental health in children and adolescents with disability: A systematic review of controlled trials. *Frontiers in Psychology. 2025;16:1567978.* DOI: https://doi.org/10.3389/fpsyg.2025.1567978

[P9] **Lee Y**, Capranica L, Pesce C, Guidotti F, Benzing V, Hauck J, Ciaccioni S. Olympic combat sports and mental health in children and adolescents with disability: A protocol paper for systematic review. *PLoS ONE. 2025;20(2):e0301949.* DOI: https://doi.org/10.1371/journal.pone.0301949

[P8] Sung E*, **Lee Y***, Kim S, Park J. (*co-first authors) Analysis of sleep duration, energy intakes, physical activity, and metabolic syndrome based on the presence or absence of obesity and hypertension in working Korean adults. Frontiers in Public Health. 2025;13:1588706. DOI: https://doi.org/10.3389/fpubh.2025.1588706

[P7] Ciaccioni S, **Lee Y**, Guidotti F, Stankovic N, Pocecco E, Izzicupo P, Capranica L. Combat sports and wellbeing: Advancing health and inclusion in athletes and practitioners. An opinion paper. *Frontiers in Psychology. 2025;16:1587672.* DOI: https://doi.org/10.3389/fpsyg.2025.1587672

[P6] Kim Y, Kim E, **Lee Y**, Park J. Role of late-night eating in circadian disruption and depression: A review of emotional health impacts. *Physical Activity and Nutrition. 2025;29(1):18-24.* DOI: https://doi.org/10.20463/pan.2025.0003

[P5] **Lee Y**, Kim N, Go S, Kim J, Park J. Sugary snack restriction enhances body composition improvement in overweight women engaging in non-face-to-face walking during COVID-19. *Frontiers in Public Health. 2024;12:1396598.* DOI: https://doi.org/10.3389/fpubh.2024.1396598

[P4] **Lee Y**, Hauck J, Brooks D. Sex-specific physical activity levels and energy intake in US young adults with depression: National Health and Nutrition Examination Survey 2017–2020 pre-pandemic data. *Physical Activity and Health. 2023;7(1):332-344.* DOI: https://doi.org/10.5334/paah.305

[P3] **Lee Y**, Park Y, Lee J, Sung E, Lee H, Park J. Household-specific physical activity levels and energy intakes according to the presence of metabolic syndrome in Korean young adults: Korean National Health and Nutrition Examination Survey 2016–2018. BMC Public Health. 2022;22:476. DOI: https://doi.org/10.1186/s12889-022-12852-3

[P2] Sung E, Lee S, **Lee Y**, Lee S, Park J. Analysis of energy intakes, physical activities and metabolic syndrome according to the income level in Korean elderly people: Korean National Health and Nutrition Examination Survey 2016–2018. *Physical Activity and Nutrition. 2022;26(2):28-35.* DOI: https://doi.org/10.20463/pan.2022.0011

[P1] Park J*, Park Y*, **Lee Y**, Lee J, Lee S, Shin C, Sung E. (*co-first authors) Comparative analysis of energy intake and physical activity according to household type and presence of metabolic syndrome in middle-aged men based on data from the 7th Korea National Health and Nutrition Examination Survey (KNHANES) (2016–2018). *Physical Activity and Nutrition. 2021;25(4):1-9.* DOI: https://doi.org/10.20463/pan.2021.0020

Manuscripts in Revision

[RV5] **Lee Y**, Vazou S, Kim S, Hauck J. Effects of a 6-Week Virtual Boxing on Depressive Symptoms and Heart Rate Variability in Adults: A Pilot Randomized Controlled Trial. *Sport Sciences for Health.* Under review (1st major revision).

[RV4] **Lee Y**, Kim N, Ciaccioni S. Wearable-Derived Behavioral Data for Depression-related/internalizing Outcomes Monitoring in Autistic Individuals: A Scoping Review. *Review Journal of Autism and Developmental Disorders.* Under review (1st major revision).

[RV3] **Lee Y**, Vazou S, Kim S, Kim N, Kim Y, Hauck J. A Systematic Review of Immersive VR Exergames for Individuals with Developmental Disabilities. *Disability and Rehabilitation: Assistive Technology.* Under review (2nd major revision).

[RV2] **Lee Y**, Chun H, Kim N, Stubbs B. Nonlinear Association Between Sedentary Time and Depression in the United States: Evidence from the 2021–2023 National Health and Nutrition Examination Survey. *Mental Health and Physical Activity.* Under review (2nd major revision).

[RV1] **Lee Y**, Brooks D, Kim S, Hauck J. Organized Sports and Psychosocial Benefits for Adolescents with Autism: Evidence from the 2022 National Survey of Children's Health. *European Journal of Adapted Physical Activity.* Under review (1st major revision).

PRESENTATIONS

Published Abstracts

[PA5] Lee J, **Lee Y**, Lin P, Luo T, Bonner S (June 2025, Poster). MoodMiles: A gamified exercise application for mental and physical well-being in young adults with all abilities. *Human-Computer Interaction International, Gothenburg, Sweden.*

[PA4] **Lee Y**, Ratan R, Vazou S, Kim S, Kim Y, Hauck J (June 2025, Poster). Exploring the effects of immersive virtual reality exercise on depressive symptoms, heart rate variability, and motivation for physical activity in young adults. *NASPSPA, Lake Tahoe, United States.*

[PA3] Brooks D, **Lee Y**, Bullard L, Hauck J, Vazou S (June 2025, Poster). Exploring contextual influences on school-based physical activity programs for youth with disabilities: A scoping review. *NASPSPA, Lake Tahoe, United States.*

[PA2] **Lee Y**, Capranica L, Pesce C, Guidotti F, Benzing V, Hauck J, Ciaccioni S (June 2024, Poster). Exploring the impact of Olympic combat sports on mental health of young individuals with disabilities: A meta-review protocol with preliminary insights. *NASPSPA, Louisiana, United States.*

[PA1] **Lee Y**, Brooks D, Hauck J (June 2023, Poster). Physical activity and energy intake in US young adults with depression: National Health and Nutrition Examination Survey 2017–2020 pre-pandemic. *NASPSPA, Toronto, Canada.*

Conference Presentations

[CP13] **Lee Y,** Nye LM, Chen PW, Mitchell JA. Wearable-Derived Sleep and Physical Activity Levels and Rhythms and Depressive Symptoms in Adults with and without Autism Spectrum Disorder. Annual Meeting of the Associated Professional Sleep Societies, LLC (APSS), *2026, Baltimore, MD.*

[CP12] **Lee Y**, Nye LM, Chen PW, Bućan M, Mitchell JA (May 2026, Poster). Wearable-Derived Sleep and Physical Activity Levels and Rhythms and Depressive Symptoms in Adults with and without Autism Spectrum Disorder. Lurie Autism Institute Annual Symposium, Philadelphia, PA, United States.

[CP11] Brooks B, **Lee Y**, Hauck J, Fisher M, West P (Feb 2026, Poster). Health and the role of employment among young adults with intellectual and developmental disabilities: Scoping review. *MSU College of Nursing Research Day, Michigan, United States.*

[CP10] Brooks B, **Lee Y**, Hauck J, Fisher M, West P (Jan 2026, Poster). Health and the role of employment among young adults with intellectual and developmental disabilities: Scoping review. *DADD, California, United States.*

[CP9] **Lee Y**, Kim S (July 2025, Poster). Synergizing physical activity and organized sports: Enhancing psychosocial well-being in adolescents with ASD in post-pandemic. *World Congress of Sociology of Sport, Seoul, Korea.*

[CP8] Kim S, **Lee Y**, Park S (July 2025, Poster). How team sports shape health in youth with autism: Examining the mediating effects of flourishing and socialization. *World Congress of Sociology of Sport, Seoul, Korea.*

[CP7] Brooks D, Hauck J, **Lee Y** (Sep 2024, Oral). Exploring physical activity experiences in children with autism spectrum disorder. *NAFAPA, New York,*

United States.

[CP6] **Lee Y**, Brooks D, Hauck J (Sep 2024, Oral). Assessing the impact of an immersive virtual reality exercise on depression and anxiety in young adults with ASD. *NAFAPA, New York, United States.*

[CP5] Ciaccioni S, **Lee Y**, Hauck J, Capranica L, Guidotti F, Benzing V, Pesce C (July 2024, Oral). Olympic combat sports and mental health in children and adolescents with disability – preliminary results of a systematic review. *ECSS, Glasgow, Scotland.*

[CP4] Anttila E, Brooks D, **Lee Y**, Hauck J (April 2023, Poster). Examining the differences in PA levels between unified partners and Special Olympic athletes at the 2022 Special Olympics Unified Cup Soccer. *UURAF, Michigan, United States.*

[CP3] Go S, **Lee Y**, Kim N, Park J (Aug 2022, Oral). The effects of non-face-to-face intervention of brisk walking and sugar restriction on body weight, physical fitness and insulin sensitivity in overweight and obese men. *KAHPERD, Gyeonggi-do, Korea.*

[CP2] **Lee Y**, Kim N, Go S, Park J (Mar 2022, Oral). Effect of non-contact brisk walking and sugar intake restriction on body fat and osteocalcin (OC) in young women during the COVID-19 outbreak period. *KSSO, Seoul, Korea.*

[CP1] **Lee Y**, Kim N, Go S, Park J (Oct 2021, Oral). Effect of non-contact brisk walking and sugar intake restriction on body fat and osteocalcin (OC) in young single-person households during the COVID-19 outbreak period. *KAHPERD, Gumi, Korea.*

Invited Lectures & Keynote Presentations

[L1] **Lee Y** (Mar 2026, Oral). Bridging the Gap: Immersive Virtual Reality as a New Pathway for Physical Activity and Mental Health in Autism. *World Autism Awareness Day Symposium, East China Normal University, Shanghai, China.*

RESEARCH GRANTS

Research Grants (Principal Investigator)

- Future of Work Graduate Student Funding, Michigan State University (2025), $3,000
- Hard Cost Practicum/Dissertation Funding, Michigan State University (2024), $1,763

Research Fellowships

- KIN Endowed Fellowship, Michigan State University (2025), $1,500
- Summer Research Fellowship, Michigan State University (2025), $7,000
- Summer Research Fellowship, Michigan State University (2024), $7,000
- Summer Research Development Fellowship, Michigan State University (2023), $7,000
- Summer Research Development Fellowship, Michigan State University (2022), $6,000

Fellowship Applications (Not Funded)

- Izaak Walton Killam Memorial Postdoctoral Research Fellowship, University of British Columbia (2026–2027 competition; proposed supervisor: Dr. Eli Puterman), CAD $60,000/year; School of Kinesiology nominee, not funded at the university level
- Eagles Autism Foundation Postdoctoral Fellowship (2026; mentor: Dr. Jonathan A. Mitchell); letter of intent not advanced to full application

RESEARCH ASSISTANTSHIPS

- Research Assistant, Development and Operation of a Standard Physical Activity Program for Children with Obesity, Seoul Metropolitan Government (2021)
- Research Assistant, Exercise Guide Video Production for the Kkumnamu Children's Health Program, Seoul Metropolitan Dongbu Hospital (2021)
- Research Assistant, Development of a Physical Activity Program Manual for Children, Seoul Metropolitan Government (2020)

TEACHING EXPERIENCE

Instructor of Record, Department of Kinesiology, Michigan State University (2022 – 2025)

- KIN 300C Coaching Soccer (Spring 2025)
- KIN 173 Foundations of Kinesiology (Spring 2023)
- KIN 121 The Healthy Lifestyle (Spring 2023)
- KIN 116B Golf II (Fall 2022)
- KIN 108R Soccer, Indoor (Fall 2022; Fall 2023, 2 sections)
- KIN 107E Tennis (Summer 2025)
- KIN 103T Distance Running (Spring 2024; Fall 2025, online asynchronous)
- KIN 103G Power Walking (Spring 2024; Fall 2025, online asynchronous)
- KIN 103D General Conditioning I (Fall 2022; Spring 2024; Fall 2024, 3 sections; Summer 2025)
- KIN 103A Aerobic Exercise I (Fall 2022; Spring 2024; Summer 2025)

Teaching Assistant, Department of Kinesiology, Michigan State University

- KIN 492/493 KINternship, online asynchronous (Fall 2023 – Fall 2025)

TEACHING GRANTS

- Inclusive Teaching Grant, Michigan State University (2025), $1,400

TEACHING AWARDS

- Grand Prize, 2019 Teaching Demonstration Competition, College of Education, Korea University (2019)
- Excellence Award, 3rd National Good Lesson Contest, College of Education, Chungbuk National University (2019)
- Special Award (Chungcheongbuk-do Superintendent of Education), 3rd National Good Lesson Contest (2019)

CERTIFICATIONS

- Secondary School Teacher Certificate, Level II (Physical Education), Ministry of Education, Korea
- Lifestyle Sports Instructor, Level 2 (Soccer), Ministry of Culture, Sports and Tourism, Korea
- Sports Management Specialist (National Technical Qualification), Ministry of Culture, Sports and Tourism, Korea

SCHOLARSHIP AWARDS

- David Cho Scholarship Award, Michigan State University (2025), $1,000
- Dr. Kenneth Hardin Graduate Student Endowed Scholarship, Michigan State University (2025), $1,500
- Henry J. and Betty Montoye Endowed Scholarship Award, Michigan State University (2024), $2,325

TRAVEL AWARDS

- Travel Fund, Department of Kinesiology, Michigan State University (2024), $600
- COGS Conference Award, Michigan State University (2024), $300
- Travel Fund, Department of Kinesiology, Michigan State University (2024), $400
- Travel Fund, Graduate School, Michigan State University (2023), $600
- Travel Fund, Department of Kinesiology, Michigan State University (2023), $400

MENTORSHIP ACTIVITIES

Mentored Undergraduate Students in Journal Club

Ronnie Aniel, Anela Panlasigue, Sathvik Suryadevara, Emma Anttila, Monique Karoub, Elaine Nevins, Brooke Johnson, Chloe Barnes, Rylee Bernhauser, Landon Moreno, Bronwen Smith, Christopher Stubbe, Laura Whitney, Maryann Hermiz, Alaina Barringer, Sarah Segarra, Lydia Miller, Morgan Adelini

PROFESSIONAL AFFILIATIONS

- Associated Professional Sleep Societies (2026 – Present)
- North American Federation of Adapted Physical Activity (2024 – Present)
- North American Society for the Psychology of Sport and Physical Activity (2023 – Present)
- Korean Society of Exercise Nutrition (2021 – Present)
- Korean Alliance for Health, Physical Education, Recreation, and Dance (2021 – Present)

SERVICE ACTIVITIES

Editorial Service

- Topic Editor, Frontiers Research Topic "Physical Education Across Hospitalization, Rehabilitation, and Return-to-School Pathways in Children and Adolescents," Frontiers in Psychology, with Dr. Simone Ciaccioni and Dr. Maria Ruberto (2026 – Present)

Scholarly and Professional Organizations

- Student Ambassador, NASPSPA, Toronto, Canada (2023)
- Editorial Director, Korean Society of Exercise Nutrition, Seoul, Korea (2022)

Reviewing for Journals

- Games for Health Journal (2026 – Present)
- American Journal of Preventive Medicine (2026 – Present)
- BMC Sports Science, Medicine and Rehabilitation (2026 – Present)
- Sport Sciences for Health (2026 – Present)
- Frontiers in Psychology (2026 – Present)
- Journal of Applied Research in Intellectual Disabilities (2025 – Present)
- Mental Health and Physical Activity (2025 – Present)
- BMJ Open (2025 – Present)
- Applied Psychology: Health and Well-Being (2025 – Present)
- Frontiers in Psychiatry (2025 – Present)
- Journal of Physical Activity and Health (2024 – Present)
- Journal for Advancing Sport Psychology In Research (2024 – Present)

COMMUNITY ENGAGEMENT AND VOLUNTEER SERVICE

- Student Mentor, Seonbae Donghaeng Mentoring Program, Munsan Middle School, Korea (2021 – 2022)
- Career Mentoring Lecturer, Sau High School, Korea (2021)
- Student Representative and Youth Sport Instructor, YOUNG MOVERS Youth Futsal and Basketball Programs, Descente Sports Foundation and Korea University (2020)
- After-School Care Volunteer, Sincheon Elementary School, Seoul (2018; 60 hours)
- Learning Mentor and Career Counselor for Children of Multicultural Families, Ulleung County Multicultural Family Support Center (2016 – 2017; volunteer award, 2017)

MILITARY SERVICE

- Ulleungdo & Dokdo Security Unit, Gyeongbuk, Korea (2016 – 2017), Mandatory Service as a Police Officer

MEDIA & PRESS COVERAGE

- WKAR Public Media (October 21, 2025). *From the ring to the MSU classroom: How VR technology is helping with autism spectrum disorder.* Link
- Korea University College of Education News (January 5, 2020). Model lesson presented at the 2019 KU Pre-service Teacher Expo after winning the Grand Prize in the 2019 Teaching Demonstration Competition.
- Gyeongbuk Maeil (November 1, 2017). Recognition for learning-mentoring volunteer work with children of multicultural families on Ulleungdo.

SKILLS

- Expertise in analyzing large-scale population health datasets (e.g., NHANES, NSCH, HBSC, KYRBS, All of Us Research Program) for physical activity epidemiology
- Experienced in processing and analyzing wearable-derived behavioral data (e.g., Fitbit, GENEActiv, ActiGraph) for physical activity, sleep, and circadian rhythm metrics
- Proficient in statistical analysis using R and SPSS; experienced in Python for simulation studies and data processing
- Experienced in designing and implementing immersive virtual reality (VR)-based exercise interventions
- Experienced in biological and physiological assessments, including blood biomarker analysis (e.g., glucose, insulin, osteocalcin) and heart

rate variability (e.g., RMSSD, SDNN, LF/HF ratio)

- Experienced in conducting systematic and scoping reviews aligned with PRISMA guidelines

Last Updated: Oct 2026